

AUTOMATED RESEARCH REPORT

Cervical Spine MRI Analysis Report

EXAM MRI cervical spine	TECHNIQUE Automated analysis of sagittal cervical spine MRI. Research-use structured interpretation only; not a substitute for radiologist review.
CASE ID scan_demo_c5	JOB ID scan_demo_c5
SUBMITTED 2026-06-07T14:10:00Z	SOURCE FILE scan.nii.gz
PATIENT CONTEXT Male, 42 years, 178 cm	SUMMARY 4 rows, 3 flagged, 1 syndromes

FINDINGS

Alignment

Cervical alignment is straightened / low lordosis. C3-C7 Cobb angle measures -2.0 deg. Approximate classification only: derived from C3-C7 Cobb on supine MRI.

Level-Specific Findings

At C5, functional canal / dural-sac AP minimum (SCT) 9.8 mm (stenosis_provisional); space available for the cord (canal AP - cord AP) 2.7 mm (high_risk); cord T2-hyperintensity screen (myelomalacia) 1.0 present (signal_anomaly_present).

Cord / Syndrome Pattern

C5: pattern consistent with possible cervical myelopathy; clinical correlation required

IMPRESSION

1. C5: pattern consistent with possible cervical myelopathy; clinical correlation required
2. C5: functional canal / dural-sac AP minimum (SCT) 9.8 mm (stenosis_provisional).
3. C5: space available for the cord (canal AP - cord AP) 2.7 mm (high_risk).
4. C5: cord T2-hyperintensity screen (myelomalacia) 1.0 present (signal_anomaly_present).
5. Alignment: straightened / low lordosis; C3-C7 Cobb angle -2.0 deg.

DISCLAIMERS

- This is a research tool, not a medical device. Not for clinical diagnosis.
- Measurements acquired on supine MRI; functional radiographs may differ.

TECHNICAL APPENDIX

Explainability Appendix

STRUCTURED FINDINGS TABLE

LEVEL	MEASUREMENT	VALUE	STATUS	QUALITY FLAGS	CAVEAT
C5	functional canal / dural-sac AP minimum (SCT)	9.8 mm	outside_reference - stenosis_provisional	None	Measures SOFT-TISSUE dural-sac AP via SCT, NOT the osseous canal.
C5	space available for the cord (canal AP - cord AP)	2.7 mm	outside_reference - high_risk	sac_slice_misaligned	Derived metric; confirm with segmentation QC.
C5	cord T2-hyperintensity screen (myelomalacia)	1.0 present	outside_reference - signal_anomaly_present	None	Cord T2-hyperintensity SCREEN (SCIseg): a positive level is a finding for physician review.
C3-C7	global cervical Cobb angle (C3-C7, lordosis-positive)	-2.0 deg	review_only - kyphotic	None	Our Cobb is C3-C7 inferior-inferior and SUPINE.

QUALITY AND CAVEATS

Measurement Notes

- C5: SAC [sac_slice_misaligned]

General Caveats

- Measures SOFT-TISSUE dural-sac AP via SCT, NOT the osseous canal.
- Derived metric; confirm with segmentation QC.
- Cord T2-hyperintensity SCREEN (SCIseg): a positive level is a finding for physician review.
- Our Cobb is C3-C7 inferior-inferior and SUPINE.

- Provisional combination rule; advisory only, never diagnostic.

THRESHOLD PROVENANCE

MEASUREMENT	TAG	CITATION	NOTE
functional canal / dural-sac AP minimum (SCT)	raw	Nell 2019 (PMC6764695, C2-C7 healthy age/sex percentiles); stenosis bands >13/10-13/<10 mm are a clinical convention (Ulbrich 2014; Thelen 2019/SHIP) of OSSEOUS-canal / radiograph origin -- pending Phase-4 MRI soft-tissue confirmation	Soft-tissue dural sac; bands provisional pending Phase-4 MRI confirmation.
space available for the cord (canal AP - cord AP)	derived	SAC <3 mm = high compression risk (plan-cited Fehlings 2015 / Nouri 2016) -- radiograph-origin, verify for MRI (pending Phase-4); Nell 2019 (PMC6764695) healthy percentiles	SAC<3 mm high-risk; radiograph-origin, verify (pending Phase-4).
cord T2-hyperintensity screen (myelomalacia)	derived	SCIseg (sct_deepseg lesion_sci_t2); Naga Karthik 2024 (PMC11065035). Measured healthy specificity 10/11 (~91%)	Group 5.1 binary screen; carried into Group 6 via the 5->6 flags contract.
global cervical Cobb angle (C3-C7, lordosis-positive)	derived	NASSJ 2025 (PMC12744292, asymptomatic C2-C7 ~18.7 deg, range ~10-35 deg); plan also cites Yukawa 2018 (~13.9 +/- 12.3 deg). Lordosis-positive.	Sign fixed (was -21 deg kyphotic); magnitude/endpoints pending SPINEPS + radiologist GT. The label side = lordosis_classification.

RAW STRUCTURED DATA

Components

```
{
  "cervical_body_morphometry": {
    "duration_s": 0.012,
    "metadata": {
      "levels": [
        "C3",
        "C4",
        "C5",
        "C6",
        "C7"
      ]
    },
    "status": "ok"
  },
  "lordosis_classification": {
    "duration_s": 0.002,
    "metadata": {
      "classification_caveat": "Approximate classification only: derived from C3-C7 Cobb on supine MRI.",
      "lordosis_classification": {
```

```
      "C3-C7": "straightened / low lordosis"
    },
    "status": "ok"
  }
}
```

Measurements

```
{
  "AP_width": {
    "C3": 19.0,
    "C4": 19.5,
    "C5": 18.9
  },
  "Cobb_C3_C7": {
    "C3-C7": -2.0
  },
  "SAC": {
    "C5": 2.7
  },
  "cord_AP": {
    "C5": 7.1
  },
  "dural_sac_AP_min": {
    "C5": 9.8
  }
}
```

Flags

```
{
  "sac_high_risk": {
    "C5": true
  },
  "sac_slice_misaligned": {
    "C5": true
  }
}
```